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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circumf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

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Day 2. Grade 12th. Questions 13, 17 and 18.



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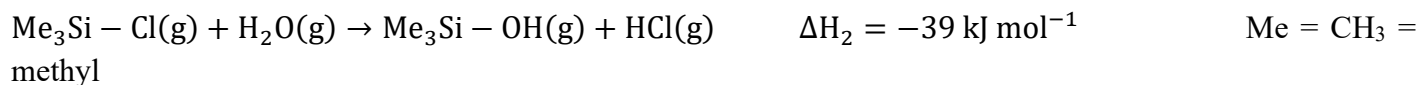
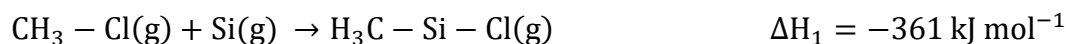
Question 13. Thermochemistry of silicon (20% of total; only 10th and 12th grades) Day 2

To the memory of Professor Dr. Jacques Rieumont Briones.

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	Total	Total
2	18	6	14	12	18	4	2	4	80	20

Silicon shares characteristics with carbon, such as being in the same family 14, forming long chains (with oxygen in silicones), forming zeolites with catalytic capacity, and having four valence states. This gives it the hypothetical ability to support life forms outside the Earth.

To study the bonds of silicon in its compounds, the heats of the following reactions in the gaseous state at very high temperatures were measured in a pump calorimeter (at constant volume), and using the First Principle of Thermodynamics, the enthalpy changes were calculated.



13.1. What thermodynamic magnitude can be determined directly in a pump calorimeter?

13.2. Calculate the bond energies E (in kJ/mol) of Si-Cl, Si-C, Si-O and C-Cl, given the values $E(\text{O-H}) = 463 \text{ kJ/mol}$, $E(\text{Cl-Cl}) = 243 \text{ kJ/mol}$ and $E(\text{Cl-H}) = 431 \text{ kJ/mol}$.

Silanes ($\text{Si}_n\text{H}_{2n+2}$) are the analogs of alkanes but derived from silicon. Silanes consist of a chain of silicon atoms bonded to hydrogen atoms. The atomization energy (ΔH_{atom}) is the energy needed to break all the bonds of 1 mole of compound, forming only atoms in the gaseous state. For silanes, ΔH_{atom} depends on the number of silicon atoms, as shown below.

Silane	Si_4H_{10}	Si_5H_{12}	Si_6H_{14}	Si_7H_{16}	Si_8H_{18}	Si_9H_{20}
ΔH_{atom} (kJ/mol)	3900	4700	6000	6500	7400	8200

13.3. Find a linear equation that relates ΔH_{atom} to the n number of silicon atoms in the formula of a silane ($\text{Si}_n\text{H}_{2n+2}$) and to the energies of the bonds Si-Si and Si-H.

13.4. Calculate $E(\text{Si-Si})$ and $E(\text{Si-H})$ by linear regression or graphically, **as accurately as possible**. Register your calculations and reasoning.

13.5. Represent the structures of silicon hydride ($n = 1$), disilane ($n = 2$), and trisilane ($n = 3$). Calculate your atomization energies.

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Methane (CH₄) is stable in air at room temperature and requires an ignition source to initiate combustion. In contrast, its silane analog is pyrophoric, meaning it spontaneously combusts upon contact with air. The structure of the reaction products also differs; while CO₂ is molecular and has two C=O bonds, SiO₂ is atomic and has four Si-O bonds per structural unit.

Enlace	C-H	C=O	O-H	O=O	Si-H	Si-O
E (kJ/mol)	413	799	467	498	318	463

13.6. Estimate the enthalpies of combustion for both substances and say which reaction is more favored.

Ethane (CH₃-CH₃) is a very stable molecule, with the C-C bond notoriously inert toward water and dilute bases. However, its disilane analog reacts easily with water or aqueous potassium hydroxide, releasing hydrogen.

13.7. Explain the difference in the reactivity of the two substances.

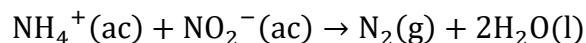
13.8. Write the balanced chemical equation for the hydrolysis reaction of disilane with hydroxide.

13.9. Trimethylamine N(CH₃)₃ has a pyramidal geometry on the nitrogen atom. However, the tris(trimethylsilyl)amine N(Si(CH₃)₃)₃ is trigonal flat. Explain the origin of the different geometries.

Question 17. Kinetics of ammonium cation (20% of total; only **12th grade**) **Day 2**

17.1	17.2	17.3	17.4	17.5	17.6	17.7	17.8	17.9	17.10	17.11	17.12	17.13	17.14	17.15	Total	Total
5	4	8	8	3	12	3	4	4	3	2	2	14	4	4	80	20

The youngest student of the last Preselection is called Cristhian, and he is passionate about Inorganic and Physical Chemistry. In the laboratory of the Faculty of Chemistry, he designed an experiment to determine the rate law for the reaction between nitrite and ammonium ions.



Cristhian took ten clean, dry containers. In five labeled containers, he added different volumes of ammonium chloride solution (5.0 mol/L); in the other five equally labeled containers, he added different volumes of sodium nitrite solution (5.0 mol/L), as indicated in the table. Then, he mixed the ammonium chloride solutions with the sodium nitrite solutions in an orderly manner, added distilled water and an inert phosphate buffer of pH = 3 to obtain a total volume of 100 mL of solution, although with different concentrations, and weighed each mixture. After 2 minutes of reaction, he weighed the mixtures again.

The difference in mass Δm for each experiment, the product of the reaction that occurred, is shown in the following table.

Exp.	Δm (g)	T (°C)	$V_0(\text{NH}_4^+)$ (mL)	$V_0(\text{NO}_2^-)$ (mL)	$V_0(\text{H}_2\text{O})$ (mL)	$V_0(\text{buffer})$ (mL)
1	0,151	25	20	20	50	10
2	0,236	25	20	25	45	10
3	0,085	25	20	15	55	10
4	0,106	25	25	15	50	10
5	0,064	25	15	15	60	10

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17.1. Prove that the above reaction occurs almost completely by calculating the equilibrium thermodynamic constant. E° (V): $\text{N}_2/\text{NH}_4^+ = -0.28$; $\text{NO}_2^-/\text{N}_2 = 1.45$.

17.2. Explain why: a) the NH_4^+ cation is a Lewis acid, b) the NO_2^- anion is an ambidentate base, c) the dinitrogen is very stable, and d) the water is amphoteric.

17.3. With the data obtained, Cristhian calculated the reaction rate for each experiment, expressed as moles of N_2 formed per liter of solution per second. What values did he obtain?

17.4. What is the expression for the rate law, rounding orders to integers?

17.5. Calculate the apparent rate constant of the reaction at 25 °C.

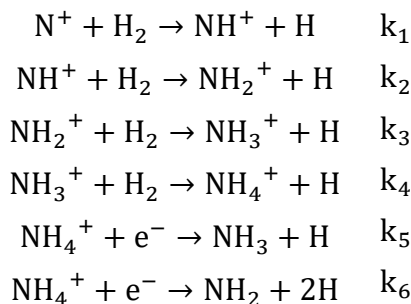
17.6. Help Cristhian to determine the activation energy (in kJ/mol) of the reaction, knowing the values of the specific rate constant for each temperature. To do this, perform a linear regression (provide the value of r) or make a graph, using the Arrhenius equation.

T (°C)	25	30	35	40	45	50
k_r (u)	0,45	0,68	1,0	1,5	2,2	3,2

17.7. It is known that the rate of this reaction depends on the concentration of protons. Using the data in the table above and the constant you calculated in **17.3**, determine the order of reaction with respect to $[\text{H}^+]$.

17.8. When Cristhian determined the total order, he realized that the reaction had to occur by a complex mechanism, which can be ionic or radical. Propose a mechanism showing the flow of electrons using arrows. Consider that H_2NNO is one of the intermediates that is formed.

Adolfo is the tallest student in the pre-selection and is very curious. After seeing the experiment above, he wondered if that reaction could happen in space. While searching the literature for data, he found that at least the ammonium cation is present in interstellar gas clouds and is involved in the following complex mechanism.



17.9. What are the boundary orbitals involved in the reaction of the first step of the above mechanism? Classify them into HOMO, LOMO, or SOMO.

17.10. The ionization energy of the nitrogen atom is 14.53 eV. Help Adolf calculate the maximum wavelength a photon can have so that it is capable of forming the N^+ ion.

17.11. In reality, the NO_2^- anion is not stable in space. Why?

17.12. Sort the N^+ , NH^+ , NH_2^+ , NH_3^+ , and NH_4^+ species with respect to their electron affinity.

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17.13. Use the steady-state approximation to derive equations for the concentrations of the intermediates NH^+ , NH_2^+ , NH_3^+ , and NH_4^+ in terms of the concentrations of $[\text{N}^+]$, $[\text{H}_2]$, and $[\text{e}^-]$ only.

17.14. Demonstrate that the rate of ammonia formation depends only on the concentration of N^+ and H_2 . Find an expression for k in terms of the velocity constants of the elementary steps.

$$\frac{d[\text{NH}_3]}{dt} = k \cdot [\text{N}^+][\text{H}_2]$$

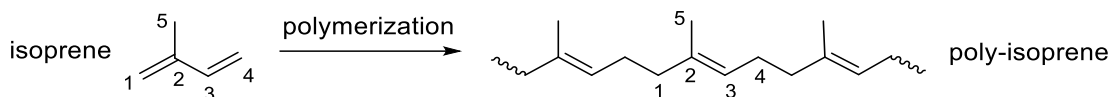
In his research, Adolfo found that the rates of many reactions involving ions and simple molecules with radicals show virtually no temperature dependence.

17.15. Considering the origin of the activation energy, explain the above fact and discuss its relevance to the reactions that occur in interstellar space.

Question 18. Luffy's skin (20% of total; only 12th grade) **Day 2**

18.1	18.2	18.3	18.4	18.5	18.6	18.7	18.8	Total	Total
8	18	6	14	10	6	10	8	80	20

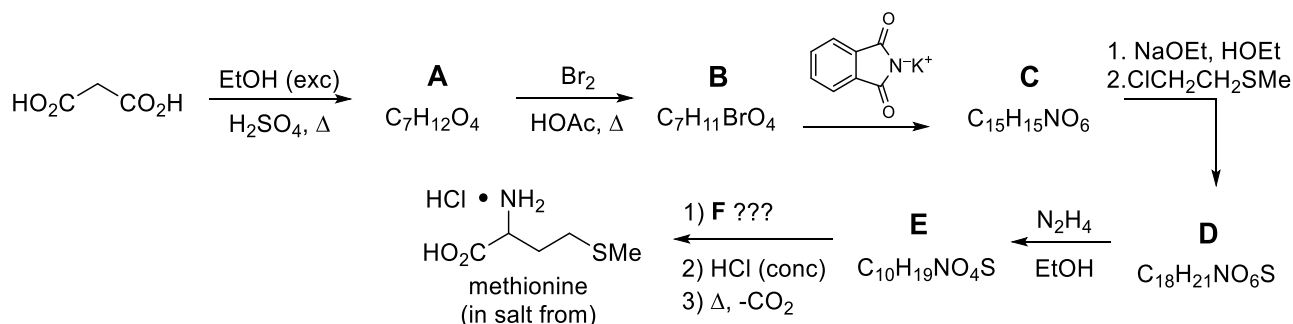
Luffy is the main character of the Japanese anime series One Piece and is very popular among students for his bravery, great strength, sense of justice, and free spirit. Luffy's skin is made of polyisoprene, the main component that gives natural rubber its elasticity.



18.1. Which of the following reagents are capable of inducing isoprene polymerization?

Reagent: a) $\text{K}_2\text{S}_2\text{O}_8$; b) H_2O ; c) BuLi ; d) BF_3 ; e) NaIO_4 ; f) $(\text{PhCO}_2)_2$; g) Zn , HCl ; h) $\text{Na}(\text{Hg})$; i) Br_2 ; j) I_2 , NaOH ; k) EtONa ; l) TiCl_4 ; m) O_3 ; n) EtOLi , Et_2Zn ; o) H_2 , Pd ; p) NaBH_4 ; q) LDA .

Luffy needs to consume moderate amounts of sulfur; so their diet should be rich in the natural amino acids cysteine and methionine, which are very abundant in foods such as meats, eggs, nuts and legumes. When food is scarce during long voyages in the Blue Sea, the crew doctor, Dr. Chopper, who is also an excellent chemist, prepares a food supplement of artificial methionine in the laboratory of the Thousand Sunny by synthesis with potassium phthalimide:



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18.2. Represent the structures of the **A-E** compounds and suggest the **F** reagent.

Luffy's metabolism is peculiar. Methionine and cysteine are enzymatically transformed into elemental sulfur, which in turn reacts with the poly-isoprene present in the skin through a process known as "vulcanization", forming C-S-C, C-S-S-C and C-S_n-C bonds between the polymer chains, and giving it greater strength and elasticity to the skin.

18.3. Name which positions (1, 2, 3, 4, or 5) of the polymer are most susceptible to the vulcanization reaction and order the three types of bonds formed during vulcanization according to their thermal stability.

However, excess sulfur can be harmful, causing Luffy's skin to become hard and brittle. In these cases, Chopper prepares a special cream containing 85% sodium stearate CH₃(CH₂)₁₆CO₂Na, an excipient or additive) and 15% Ni-Raney (an active component). The salt is obtained by saponification of natural triglycerides. Ni-Raney is obtained by reaction of an equimolar alloy (1:1) of nickel and aluminum with concentrated and hot sodium hydroxide, until all the aluminum is consumed and only chemisorbed hydrogen remains onto finely divided nickel (of NiH_x composition, 1 < x < 2).

18.4. What minimum masses of nickel, aluminum, sodium hydroxide and stearate are needed to prepare one kilogram of special Chopper cream? Suppose that the Ni-Raney has NiH_{1.5} composition and that the aluminum in the alloy is dissolved in the form of [Al(OH)₄]⁻.**18.5.** During the skin treatment with special cream, sailor Nami and cook Sanji are always in a bad mood, because the whole boat takes on a very unpleasant smell. Formulate the equations for the chemical reactions that occur, given that a black solid is one of the products. Suppose the polymer formula of Luffy's skin is R-S_n-R when fully vulcanized.

During the assault on Impel Down Prison to rescue his brother, Luffy was attacked by the head manager, Magellan, whose body is covered in highly toxic substances such as Br₂, H₂SO₄ and HNO₃, concentrated and hot. Fortunately, Luffy's immune system produces antidotes for these poisons, which facilitate their elimination, for example, p-hydroxybenzoic acid.

18.6. Classify the hydroxyl and carboxylic acid groups according to their electronic effect on the S_EAr reaction. Justify your answer by the resonant structures of phenol and benzoic acid.**18.7.** Represent the structures of the main reaction products of p-hydroxybenzoic acid with Br₂, H₂SO₄ and HNO₃, separately. In the reaction with a halogen, a double substitution occurs.**18.8.** What are the true electrophiles in S_EAr reactions with H₂SO₄ and HNO₃? Write the chemical equation for the electrophile formation reaction in each case.

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